

Harmonis Prime: A Reproducible Distributed Systems Benchmark Specification (HBS-1.1)

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Abstract

We present Harmonis Prime, an experimental distributed systems stack implemented in Rust, integrating Raft consensus, deterministic chaos injection, formal verification primitives, quantum-classical simulation, and a reproducible benchmark harness under a unified architecture. The system is validated by 50 passing unit and integration tests, continuous integration across three operating systems (Linux, Windows, macOS), and one independent external reproduction on commodity hardware by an unaffiliated engineer (@abdulwahab72). All benchmarks execute with a fixed pseudorandom seed (0x51C3_2026_0613) to guarantee deterministic operation sequences, while wall-clock timing is reported with honest variance bounds ($\pm 30\%$) reflecting real-world consumer hardware conditions. This paper documents the architecture, benchmark specification, validation methodology, and honest limitations of the current release, establishing a reproducibility baseline for future iterations toward thermodynamic efficiency claims.

Keywords: distributed systems, reproducible benchmarking, Raft consensus, Rust, deterministic chaos injection, formal verification, quantum-classical simulation

1. Introduction

1.1 Motivation

The field of distributed systems suffers from a reproducibility gap. Benchmark claims are frequently made without corresponding open-source implementations, deterministic harnesses, or independent validation. Performance numbers are cited without hardware context, energy measurement, or statistical rigor. This creates an environment where claims outpace measurements, and trust erodes.

Harmonis Prime was built to close this gap. The governing principle is simple:

Claims = Measurements. Nothing more. Nothing less.

Every architectural claim is backed by code. Every performance claim is backed by a deterministic benchmark. Every validation claim is backed by tests, CI, and human witness logs.

1.2 Contribution

This release (v7.1.1, HBS-1.1) contributes:

A unified stack where transport, consensus, storage, simulation, and verification layers coexist under a single reproducible harness.

Deterministic benchmarks with fixed seeds, honest hardware context, and documented variance.

Multi-platform validation via GitHub Actions CI (Linux, Windows, macOS).

External human witness validation with full reproduction logs.

Honest limitation documentation — every non-claim is explicitly listed.

1.3 Scope

This paper covers the HBS-1.1 specification. It does not claim physical quantum integration, energy efficiency superiority, or production-grade mesh networking. Those are future milestones on the 12-month summit roadmap.

2. Architecture

Harmonis Prime is organized into six primary layers, each corresponding to a BRICK module in the source tree.

2.1 Quantum-Classical Simulation Layer (src/brick42/quantum/)

A simulated quantum substrate providing amplitude estimation and normalization. This is a simulated backend only — no physical QPU integration exists in HBS-1.1. The layer serves as a placeholder for future quantum-classical hybrid computation, with classical emulation of quantum state transitions.

Status: Implemented (simulated).

Future: Physical QPU integration planned for HBS-1.3+.

2.2 Formal Verification Engine (src/fv/)

An invariant checker, model checker, and proof generator operating over the system state space. Validates causal consistency and policy compliance at runtime.

Status: Implemented.

Coverage: BRICK-45 through BRICK-51 invariants.

2.3 Policy-Driven Autonomy (src/autonomy/)

A predicate calculus runtime enabling declarative policy specification over system behavior. Policies are evaluated against the formal verification engine before state transitions are committed.

Status: Implemented.

2.4 Causal State Transitions (src/brick49/)

Every state transition produces a `CausalityProof` structure documenting the predecessor state, operation, and resulting state hash. This forms an immutable causal chain.

Status: Implemented.

2.5 Raft Consensus (src/raft/)

A Raft implementation [7] supporting leader election, log replication, and deterministic chaos injection. Chaos scenarios simulate network partitions, leader failures, and message delays to validate consensus robustness.

Status: Implemented.

Tests: 4 cluster-level integration tests pass.

2.6 Shared-Memory Graph (src/brick51/)

A single-node `HashMap`-backed graph abstraction providing the storage substrate for Workload A. This is single-threaded simulation — no physical cluster or distributed memory is used in HBS-1.1.

Status: Implemented.

Future: DAG-enforced mesh topology planned for HBS-1.2.

3. Benchmark Specification

3.1 Design Principles

All benchmarks follow these principles:

Determinism: Fixed PRNG seed guarantees identical operation sequences across runs.

Honesty: Wall-clock timing is reported with explicit variance bounds and hardware context.

Independence: Zero external benchmark dependencies — only std and project-internal crates.

Reproducibility: A single cargo test invocation reproduces all validation.

3.2 Workload A: SharedMemoryGraph Microbenchmark

Table

Attribute Value

Name `harmonis_shared_memory_graph`

Operation SharedMemoryGraph::insert + SharedMemoryGraph::get

Node config node_id=0, node_count=1

Determinism Fixed seed: 0x51C3_2026_0613

Iterations 10,000

Metric Per-iteration wall-clock latency (ns)

3.3 Workload B: Consensus Simulation

Table

Attribute Value

Name harmonis_consensus_simulation

Operation Deterministic PRNG + chaos scenario + Raft leader election + heartbeat + consistency check

Node config 5 simulated nodes, 7 chaos scenarios

Determinism Fixed seed: 0x51C3_2026_0613

Iterations 10,000

Metric Per-iteration wall-clock latency (ns)

Honest Label SIMULATION — production APIs not yet exposed

3.4 Build Specification

Table

Attribute Value

Language Rust 1.96.0 (pinned via rust-toolchain.toml)

Profile release (optimized)

Target x86_64-pc-windows-msvc (primary)

Dependencies Zero external benchmark dependencies

3.5 Hardware Context (Developer Baseline)

Table

Attribute Value

Machine Consumer laptop, stock configuration

CPU 11th Gen Intel i7-1165G7

RAM 16GB DDR4

OS Windows 11

Core pinning NOT IMPLEMENTED

Turbo locking NOT IMPLEMENTED

Energy measurement NOT AVAILABLE

3.6 Timing Methodology & Statistical Reporting

Table

Attribute Value

Timer std::time::Instant::now() (OS wall-clock)

Variance source Turbo Boost, OS scheduling, thermal throttling

Expected run-to-run variance $\pm 30\%$ on consumer hardware

Statistical method Single-run reporting (median-of-N planned for HBS-1.2)

Workload determinism  Fixed seed guarantees identical operations

Measurement determinism  Wall-clock timing varies with system state

Honest framing: This benchmark measures real-world latency under real-world conditions. It does NOT claim cycle-accurate reproducibility. For that, core pinning + rdtsc would be required.

4. Validation & Reproducibility

4.1 Self-Verification

Table

Environment OS Result Witness

Developer machine Windows 11  50 pass, ~10 min Self

4.2 CI Verification

Table

Environment OS Result Witness

GitHub Actions Ubuntu Latest  50 pass CI (Automated)

GitHub Actions Windows Server  50 pass CI (Automated)

GitHub Actions macOS Latest  50 pass CI (Automated)

CI workflow: `.github/workflows/rust.yml` (commit 374bd6a).

Trigger: Every push to main and every pull request.

4.3 External Human Witness

Table

Attribute Value

Witness @abdulwahab72 (independent engineer, unaffiliated)

Date 2026-06-09

Machine Windows 10 Pro

CPU Intel Core i7-8565U (8th Gen)

RAM Not specified

OS Windows 10 Pro

Rust Version 1.96.0

Tag Tested v7.1.1-SPEC-HBS1.1

Commit 14f1de0

Commands Executed:

bash

git clone https://github.com/Ayub19123/Harmonis-Prime.git

cd Harmonis-Prime

git checkout v7.1.1-SPEC-HBS1.1

cargo test

cargo audit

cargo run --release --bin benchmark -- 10000 0x51C3_2026_0613

Results:

Table

Check Result

cargo test  PASSED — 0 failures

cargo audit  PASSED — 0 vulnerabilities

Benchmark  PASSED — deterministic output with fixed seed

Runtime 1100.32s (~18 minutes)

4.4 Hardware Variance Analysis

Table

Environment CPU Runtime Variance

Baseline (Developer) Intel i7-1165G7 (11th Gen) ~10 minutes Baseline

Witness #1 (Abdulwahab72) Intel i7-8565U (8th Gen) ~18 minutes +80%

GitHub Actions (CI) VM (3 OS) ~12 minutes +20%

Conclusion: Runtime variance is consistent with documented $\pm 30\%$ wall-clock variance on consumer hardware. The ~80% increase on i7-8565U vs i7-1165G7 is explained by generational CPU differences (8th Gen vs 11th Gen). The deterministic operation sequences remain identical; only wall-clock timing varies.

4.5 Validation Matrix Summary

Table

Environment OS Status Witness

Developer machine Windows 11  50 pass Self

GitHub Actions Ubuntu  50 pass CI

GitHub Actions Windows Server  50 pass CI

GitHub Actions macOS  50 pass CI

External witness Windows 10  50 pass Abdulwahab72

Total independent validations: 5 environments, 1 human witness.

4.6 Reproduction Protocol

To reproduce this validation:

```
bash
```

```
git clone https://github.com/Ayub19123/Harmonis-Prime.git
```

```
cd Harmonis-Prime
```

```
git checkout v7.1.1-SPEC-HBS1.1
```

```
cargo test # ~10-20 minutes, 50 tests
```

```
cargo audit # 0 vulnerabilities expected
```

```
cargo run --release --bin benchmark -- 10000 0x51C3_2026_0613
```

Expected result: All tests pass. Benchmark produces deterministic output. Runtime varies by hardware generation.

5. Honest Limitations

The following table documents every claim that Harmonis Prime HBS-1.1 does not make:

Table

Claim Reality

Quantum substrate Simulated backend only — no physical QPU integration

Density matrix evolution Amplitude estimation with normalization — not full von Neumann equation

Mesh topology Graph abstraction exists — DAG enforcement is future work (HBS-1.2)

Hardware SIMULATED on consumer laptop, stock config — no physical cluster

Energy NOT MEASURED — no power profiling available

Multi-node Single-threaded simulation — no physical cluster

Statistical rigor Single-run reporting — median-of-N planned for HBS-1.2

Cycle accuracy Wall-clock Instant::now() — no rdtsc or core pinning

Production APIs Benchmark is simulation — production APIs not yet exposed

3+ witnesses Only 1 human witness to date — more needed for statistical confidence

6. Future Work & 12-Month Summit Roadmap

6.1 HBS-1.2 (Months 3–4)

DAG-enforced mesh topology

Median-of-N statistical reporting

2nd and 3rd human witness accumulation

Rust Users Forum peer review engagement

6.2 HBS-1.3 (Months 5–6)

SET-3 thermodynamic validation protocol

Energy measurement instrumentation

Data efficiency benchmarks vs. standard transformers

Entropy reduction metrics

6.3 Summit Phase (Months 7–12)

Physical multi-node cluster (3+ nodes)

Cross-node consensus at scale (5+ nodes)

3–5 independent human witnesses

Whitepaper peer review and citation

Public launch across engineering channels

6.4 Long-Term Technical Objective

The long-term research objective is a distributed systems architecture targeting orders-of-magnitude reductions in data movement and energy consumption through deterministic execution and low-entropy state transitions. This requires thermodynamic efficiency proofs (SET-3), minimal data movement architectures, low-entropy consensus protocols, and deterministic, reproducible execution. HBS-1.1 is the first reproducible baseline on that path.

7. Conclusion

Harmonis Prime v7.1.1 (HBS-1.1) establishes a reproducible baseline for distributed systems validation. With 50 passing tests, 5 validated environments, 1 independent human witness, deterministic benchmarks, and honest limitation documentation, it demonstrates that claims can equal measurements when discipline is applied.

The system is not complete. It is not the summit. It is the first step toward a larger vision of efficient, verifiable, distributed intelligence.

Claims = Measurements. Nothing more. Nothing less.

References

Harmonis Prime Repository: <https://github.com/Ayub19123/Harmonis-Prime>

HBS-1.1 Specification Tag: v7.1.1-SPEC-HBS1.1 (commit 14f1de0)

CI Badge Tag: v7.1.1-README-CI-BADGE (commit a35f630)

Witness Tag: v7.1.1-WITNESS-1 (commit b5e61eb)

REPRODUCTION_LOG.md: Witness #1 log (commit 308416a)

GitHub Actions Workflow: [.github/workflows/rust.yml](#) (commit 374bd6a)

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Next Review: Upon accumulation of Witness #2

Maintainer: Harmonis Prime Core Team